Correlation Analysis: DEXA vs InBody 570 / InBody 270

Machines: CF597, CF661, CF625 | Overall correlation (filename): r = 0.987

InBody 270

	CF597		CF661		CF625	
Metric	Fat Mass (95% ND)	Skeletal Muscle Mass (95% ND)	Fat Mass (95% ND)	Skeletal Muscle Mass (95% ND)	Fat Mass (95% ND)	Skeletal Muscle Mass (95% ND)
Correlation coefficient	0.993	0.9965	0.9926	0.9957	0.9872	0.9962
P-value	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001
R value	0.986	0.9929	0.9852	0.9914	0.9746	0.9925
LoA	0.68±1.33	0.52±0.89	0.75±1.32	0.59±0.90	1.47±1.79	1.13±1.01

InBody 570

	CF597		CF661		CF625	
Metric	Fat Mass (95% ND)	Skeletal Muscle Mass (95% ND)	Fat Mass (95% ND)	Skeletal Muscle Mass (95% ND)	Fat Mass (95% ND)	Skeletal Muscle Mass (95% ND)
Correlation coefficient	0.9919	0.9964	0.9914	0.9943	0.9903	0.9956
P-value	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001
R value	0.9839	0.9929	0.9828	0.9886	0.9808	0.9911
LoA	0.75±1.37	0.52±0.89	1.03±1.56	0.77±1.07	1.74±1.56	1.34±1.07

The University of Hong Kong–Shenzhen Hospital (HKU-SZ Hospital)

	CF597		CF661		CF625	
Metric	Fat Mass (95% ND)	Muscle Mass (95% ND)	Fat Mass (95% ND)	Muscle Mass (95% ND)	Fat Mass (95% ND)	Muscle Mass (95% ND)
Correlation coefficient	0.9893	0.9955	0.9894	0.995	0.9901	0.993
P-value	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001
R value	0.9788	0.9911	0.9789	0.99	0.9802	0.9874
LoA	1.34±1.41	1.51±2.32	1.03±1.23	1.56±2.02	0.62±1.29	1.05±2.16

Shenzhen University General Hospital (SZU General Hospital)

	CF597		CF661		CF625	
Metric	Fat Mass (95% ND)	Muscle Mass (95% ND)	Fat Mass (95% ND)	Muscle Mass (95% ND)	Fat Mass (95% ND)	Muscle Mass (95% ND)
Correlation coefficient	0.9897	0.9939	0.9895	0.9941	0.9889	0.9945
P-value	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001	>0.0001
R value	0.9795	0.9878	0.979	0.9882	0.9779	0.989
LoA	1.44±1.51	0.95±1.6	1.21±1.44	0.84±1.6	0.81±1.55	0.95±1.49

Notes

Notes on Chart-Related Parameters:

1. Correlation: Indicates an extremely strong positive correlation between the two devices' measurements, with a highly significant linear relationship.
2. P-value: Far below the significance level of 0.05, indicating that the correlation is highly statistically significant and not due to random error.
3. Coefficient of determination (R^2): Shows the percentage of variation in fat mass (close to 1) that can be explained by the InBody 270 measurements, indicating an excellent fit.
4. Fitted formula: The fitted regression equation is $y = 0.9688x + 0.5539$. The slope is close to 1 and the intercept close to 0, suggesting that the fat-mass values measured by CF597 and InBody 270 are highly close, with a good linear relationship (the closer to 1 = the two devices change completely in sync).
5. 95% Limits of Agreement (LoA = –0.65 (mean bias) to 2.01 (max deviation)):

   Meaning: There is a 95% probability that the fat mass measured by CF597 will not exceed that of InBody 270 by more than 2.01 kg;

   Lower limit –0.65: In a few cases, CF597 measures 0.65 units less than InBody 270.

   Upper limit 2.01: In a few cases, CF597 measures 2.01 units more than InBody 270.

   Ideal case: The bias is close to 0 and the limits of agreement are narrow; fitting at 0.95 consistency indicates that the overall agreement between the two devices is good.

   ±1.96 standard deviations (95% normal-distribution lookup value) → exactly captures 95% of the data.